Atharv Ved Rai

AI Engineer — Python — Machine Learning — Deep Learning — Generative AI — LLMs

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SUMMARY

I have built numerous projects, done technical internship, which have helped me develop experience in AI and deploying models. I am a 2026 Computer Science and Engineering graduate from Vellore Institute of Technology with a specialisation in AI & ML. I specialise in building innovative solutions powered by NLP, ML, and Gen AI. I want to provide solutions to problems humanity faces through tech.

EDUCATION

B. Tech in Computer Science and Engineering with AI & ML, 2022 – 2026
Vellore Institute of Technology

SKILLS

Languages — Python, SQL, JavaScript, TypeScript

Machine Learning — Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Supervised & Unsupervised Learning, Model Evaluation

Generative AI & LLMs — LangChain, LangGraph, Hugging Face, OpenAI API, Prompt Engineering, RAG, AI Agents, Fine-Tuning

Deep Learning & NLP — ANN, CNN, RNN, LSTM, Transformers, Natural Language Processing, Text Classification, Sentiment Analysis

Frameworks & Databases — FastAPI, Flask, REST APIs, MySQL, MongoDB

Cloud & Developer Tools — AWS, Docker, Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Postman, n8n

Core Concepts — Predictive Modeling, Classification, Regression, Clustering, Data Preprocessing, EDA, Model Deployment, AI Automation

EXPERIENCE

Artificial Intelligence Intern, *FactSomeTV* 01/2025 – 03/2025
- Architect and develop AI-driven solutions across Generative AI, AI Agents, Automation, and Cloud AI and use cases into scalable technical architectures and solution strategies.
- Design and evaluate end-to-end automation solutions by analyzing workflows, identifying optimization opportunities, and recommending appropriate AI, software, and cloud technologies to improve efficiency, scalability, and system performance.
- Develop technical proposals, solution architectures, and executive presentations for AI and digital transformation initiatives.

Lab Intern, *India Space Research Organisation* 12/2024 – 01/2024
- Completed aerospace training across UAVs, CubeSats, CanSats, CFD, FEA, communication protocols, and sensor integration.
- Ran Digital Twin simulations for aircraft and pinpointed 3 critical propulsion failure points for engineering optimization.

PROJECTS

Moderna mRNA Vaccine Clinical Safety & Dosage Intelligence Platform,, *Python, Scikit-learn, Pandas, NumPy, XGBoost, Streamlit*
- Developed a clinical intelligence platform to analyze vaccine safety dataset and identify pattern in event and patient responses.
- Implemented ML models for adverse event risk classification and dosage-related data analysis using clinical features.
- Designed an interactive dashboard for clinical data visualization and risk assessment, targeting 90% classification accuracy with model evaluation and performance metrics.

FlowCast – AI-Powered Smart Traffic Flow Prediction Platform, *Python, TensorFlow, Pandas, NumPy, LSTM, Streamlit*
- Developed an AI-powered traffic forecasting platform using historical traffic data to predict congestion patterns and traffic density across different time intervals.
- Implemented LSTM-based time-series forecasting with features to analyze traffic patterns and improve prediction performance.
- Built an interactive dashboard for traffic visualization and congestion forecasting, targeting 90% prediction accuracy and reducing forecasting error by 15% against baseline models.

Vantara Customer Intelligence Platform, *Python, Scikit-learn, Pandas, SQL, Streamlit*
- Developed an AI-powered customer intelligence platform leveraging customer segmentation and predictive analytics to identifybehavioural patterns across customer datasets.
- Implemented K-Means clustering and customer churn prediction models, enabling customer profiling and retention analysis.
- Built an interactive analytics dashboard to visualize customer segments, behavioural trends, and predictive insights, targeting 85%+ model accuracy.

CERTIFICATES

- Applied Machine Learning in Python – Coursera
- AI Fluency: Framework & Foundations – Anthropic
- Foundations of Prompt Engineering – AWS

ORGANISATIONS

Media Lead, *FinTech Club* 01/2023 – 11/2025
Led coordination for 6 college events on topics like cryptocurrencies, paper trading and digital payments. With 100+ participants per event. Mentored junior members and facilitated communication between departments & faculty members.